

Aryan Singh

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computing and Financial Management (Co-op) | GPA: 3.70 / 4.00

Expected Apr. 2030

- **Relevant Coursework:** Algorithms & Data Structures, Financial Markets, Data Analytics, Financial Reporting

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL (MySQL), HTML/CSS

Frontend: React, Next.js, Tailwind CSS, Recharts, Vite

Backend & Data: Node.js, FastAPI, pandas, NumPy, Matplotlib, yfinance, scikit-learn, Streamlit

Developer Tools: Git/GitHub, Linux, VS Code, Jupyter Notebook, MS Excel, Power BI

PROJECTS

EPS Drift Scanner | Python, JavaScript, FastAPI, React, pandas, scikit-learn, Recharts

[GitHub](#)

- Built a **live EPS surprise scanner** ingesting analyst estimates vs. actuals for 10+ equities via Financial Modeling Prep API, deployed full-stack on Vercel and Render.
- Engineered a **quantitative drift pipeline** in pandas classifying quarterly EPS surprises across 5 tiers and 12 quarters with configurable Wall Street estimate alert thresholds.
- Detected anomalous earnings surprises using **scikit-learn LinearRegression** residual scoring per equity, surfacing anomaly scores via an interactive React and Recharts dashboard.

Market Stress & Crisis Simulator | Next.js, TypeScript, React, Tailwind CSS, Recharts

[GitHub](#)

- Built a **portfolio simulation engine** supporting asset weighting, normalization, and configurable rebalancing across 10+ assets and 3 major market crises.
- Engineered a **stress-testing framework** replaying 2008 GFC, COVID-19, and 2022 Rate Shock over 12–36 month horizons to simulate drawdown and recovery trajectories.
- Designed an **analytics pipeline** processing 10+ years of market data to compute 6+ risk metrics including maximum drawdown, Sharpe ratio, and time-to-recovery.

Robo-Advising Portfolio Optimizer | Python, pandas, NumPy, yfinance

[GitHub](#)

- Built a **portfolio construction pipeline** using pandas and NumPy to filter equities by currency, liquidity, and market capitalization, removing delisted tickers and handling cross-listed duplicates.
- Screened equities by **annualized volatility** to exclude high-risk equities per sector while preserving diversification across 10+ industries.
- Constructed a **\$1,000,000 CAD market-meet portfolio**, selecting equities correlated with the S&P 500 and TSX Composite and converting portfolio weights into shares, accounting for FX conversion and transaction costs.

Equity Trend Analyzer | Python, pandas, NumPy, yfinance, Matplotlib, Streamlit

[GitHub](#)

- Developed a **quantitative equity analysis tool** leveraging historical market data to compute performance and risk metrics, including maximum drawdown, total return, and annualized volatility.
- Classified trend direction (uptrend/downtrend/neutral) across **50+ equities** using linear regression on log-price data to identify momentum shifts.
- Engineered **RSI (14)** and moving average crossover signals with dynamic Matplotlib visualizations and CSV export for downstream analysis.

EXPERIENCE

Food & Beverage Service Associate

Mar. 2023 – Aug. 2025

Canada's Wonderland

Vaughan, ON

- Executed **300+ high-volume financial transactions** per shift using Oracle POS systems, maintaining 95–99% accuracy during peak operating hours.
- Coordinated operations across **5–10 associates per shift**, delegating tasks to reduce wait times and maintain service standards during peak periods.

International Service Project (Global Development)

Mar. 2024

Shelter Them Poverty Relief (NGO) — Brampton Christian School

Kigali, Rwanda

- Planned and executed fundraising initiatives, raising **\$2,000+** and coordinating budget allocation to support on-time project delivery.
- Collaborated within a cross-functional team of **15+ members** to deliver a community infrastructure project on time despite fixed timelines and logistical constraints.